

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY
(Deemed to be University)

M.Tech / Ph.D MID-III EXAMINATIONS : 08/Dec/ 2022

Course (Subject) : **Advanced Optimization Techniques /** Course Code **AM-603 /**
Computer Oriented Optimization Techniques **AM-603D**

Duration : **01 hrs** Max Marks : **10**

1. Discuss for maxima and minima for $f(x) = x^5 - 5x^4 + 5x^3 - 10$

✓ 2. Solve the following problem using Lagrangian multipliers method.

$$\text{Min } Z = x_1^2 + x_2^2 + x_3^2 - 10x_1 - 6x_2 - 4x_3$$

$$\text{S.to. } x_1 + x_2 + x_3 = 7$$

$$x_1, x_2, x_3 \geq 0$$

3. Minimize $f(x) = x^2 - 2.6x + 2$ in the $[-2, 3]$ by Fibonacci method with $N = 8$.